CASE STUDY · POWER BI · SOCIOECONOMIC ANALYSIS

Boston Liveability Dashboard

Neighborhood-level socioeconomic analysis across 41 Boston zip codes — built with US Census / ACS data, normalized in Excel, and visualized in Power BI.

May 2026

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01 OVERVIEW

Built a Power BI dashboard analyzing liveability across 41 Boston-area zip codes using US Census / ACS 5-year estimates. Data was collected via the Census API — one row per zip code — processed through a min-max normalization pipeline in Excel, and assembled into an interactive multi-metric dashboard to surface neighborhood-level insights.

02 DATA AND DIMENSIONS

Source: US Census / ACS 5-year estimates, collected via API. 41 zip codes. Five socioeconomic dimensions:

Dimension	Raw Column	Direction
Median Household Income	Median Household income	Higher = better
Education Attainment	% of bachelor degree holders or higher	Higher = better
Unemployment Rate	Unemployment Percent	Lower = better (inverted)
Median Home Value	Median home value	Higher = better
Population Density	Population Density (per sq mi)	Lower = better (inverted)

03 NORMALIZATION FORMULAS

All five dimensions were normalized to a 0–1 scale using min-max normalization so they could be combined into a single composite score despite different units and ranges.

Standard normalization (higher = better)

```
Normalized = (x - min(x)) / (max(x) - min(x))
```

```
Applied to: Median Household Income

Education Attainment (% bachelor+)

Median Home Value
```

Inverted normalization (lower = better)

```
Normalized = 1 - (x - min(x)) / (max(x) - min(x))


Applied to: Unemployment Rate

Population Density
```

A score of 1.0 means best performance on that dimension. A score of 0.0 means worst. For unemployment and density, a lower raw value yields a higher normalized score.

04 COMPOSITE SCORE FORMULA

Each zip code receives a single liveability score as a weighted sum of its five normalized dimensions. Weights reflect the relative impact of each factor on day-to-day quality of life:

```
Composite = (0.30 x Income_norm)

+ (0.25 x Unemployment_norm)

+ (0.20 x Education_norm)

+ (0.15 x HomeValue_norm)

+ (0.10 x Density_norm)
```

Dimension	Weight	Rationale
Median Household Income	30%	Strongest single predictor of financial stability and access

Unemployment Rate	25%	Reflects economic health and opportunity in the area
Education Attainment	20%	Proxy for community resources, schools, and workforce quality
Median Home Value	15%	Signals desirability but can diverge from income (South End)
Population Density	10%	Lower density loosely linked to space, noise, congestion

05 SAMPLE CALCULATION — ZIP 02199

Financial / Government / Corporate (PO Boxes) — highest composite score in the dataset at 0.837:

Dimension	Raw Value	Normalized	Weight	Contribution
Median Household Income	$106,182	0.5866	0.30	0.1760
Unemployment Rate	0.00%	1.0000	0.25	0.2500
Education Attainment	91.75%	0.9442	0.20	0.1888
Median Home Value	$2,000,001	1.0000	0.15	0.1500
Population Density	21,653/sqmi	0.7249	0.10	0.0725
			Total	0.8373

06 KEY FINDINGS

Finding	Detail
Cambridge / Downtown corridor leads	Consistently top scores across income, education, and unemployment
Roxbury, Mattapan, Dorchester bottom out	Driven by income and unemployment gaps, not density
Home value and liveability diverge	South End: $946K median home value, 9.1% unemployment, below-median income
Fenway / Back Bay unemployment anomaly	11% unemployment despite affluent surroundings — large student population distorts the metric
0% unemployment outlier	Zip 02199 is PO Boxes / corporate — no residents to be unemployed, inflates its composite score

07 DASHBOARD FEATURES

Interactive Power BI report with TomTom basemap:

Feature	Description
Choropleth map	Composite score and individual metrics shaded by zip code
Ranked bar charts	All 5 dimensions ranked across all 41 neighborhoods
Variable selector	Toggle between Education, Income, Home Value, Density, Unemployment on one map
Outlier annotations	Neighborhoods where metrics diverge from expectations flagged inline

08 TOOLS

Power BI · Excel (min-max normalization pipeline) · US Census API · ACS 5-year estimates

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